

Here if you think AI is really bad on the environment

Posted by u/ClothesPristine7428 • • 0 points • 107 comments

TLDR;

Digital art (4 hr): $\approx 22\text{-}60$ L water, $\approx 1.1\text{-}2.2$ kWh

AI image: $\approx 0.002\text{-}0.06$ L (2-60 mL) water, $\approx 0.001\text{-}0.013$ kWh

Watercolor: $\approx 0.2\text{-}1.3$ L water, $\approx 0.02\text{-}0.06$ kWh

Pencil sketch: $\approx 0.05\text{-}0.3$ L water, $\approx 0.005\text{-}0.01$ kWh

A. Small watercolor painting

Electricity (materials + making): Paper manufacturing $\approx 0.7\text{-}1.5$ kWh/kg; one A5 sheet (~ 5 g) $\approx 0.004\text{-}0.008$ kWh. Pigment/paint manufacturing adds $\sim 0.01\text{-}0.05$ kWh for the fraction of a pan/tube used. Brushes add a negligible share per painting. Total $\approx 0.02\text{-}0.06$ kWh.

Water (materials + making): Paper industry uses $\approx 10\text{-}50$ L/kg finished paper; one A5 sheet $\sim 0.05\text{-}0.25$ L. Pigment/paint production (water-based) adds $\approx 0.01\text{-}0.05$ L per painting. Artist use while painting adds $\approx 0.1\text{-}1.0$ L. Total $\approx 0.2\text{-}1.3$ L.

B. Pencil sketch

Electricity (materials + making): Graphite pencil production $\approx 0.2\text{-}0.5$ kWh/kg finished pencils; a single pencil (~ 6 g) $\approx 0.001\text{-}0.003$ kWh. Paper (A5, 5 g) adds $0.004\text{-}0.008$ kWh. Total $\approx 0.005\text{-}0.01$ kWh.

Water (materials + making): Wood + graphite pencil $\approx 0.5\text{-}1.0$ L/kg; a pencil $\approx 0.003\text{-}0.006$ L. Paper sheet $\approx 0.05\text{-}0.25$ L. No significant water use in sketching itself. Total $\approx 0.05\text{-}0.3$ L.

C. Digital art piece (4 hours on desktop)

Electricity (operation): 150-300 W desktop + 30-50 W monitor = 180-350 W. Four hours = 0.72-1.4 kWh. Tablet negligible (<0.02 kWh).

Electricity (device manufacture share): Life-cycle assessment for laptops/desktops ≈ 1,000-2,000 kWh total embedded energy; allocating 4 hours out of a 3-5 year, ~5,000-10,000 hour lifetime ≈ 0.4-0.8 kWh per 4 hours. Total ≈ 1.1-2.2 kWh.

Water (operation): Convert 1.1-2.2 kWh × 1.9-4.5 L/kWh ≈ 2-10 L.

Water (device manufacture share): Electronics manufacturing averages ≈ 50-60 L per kWh of embodied energy → ≈ 20-50 L allocated. Total ≈ 22-60 L.

D. AI-generated image

Electricity (inference only): ≈ 0.00009-0.003 kWh per image.

Electricity (training amortized): Large model training may consume millions of kWh (e.g., GPT-3 training ≈ 1,287 MWh). If amortized across billions of images, per-image share ≈ 0.001-0.01 kWh. Combined ≈ 0.001-0.013 kWh.

Water (inference only): 0.02-14 mL depending on water-use intensity.

Water (training amortized): Using U.S. average 1.9-4.5 L/kWh, training adds ≈ 2-45 mL per image. Total ≈ 2-60 mL.

Comments

u/mf99k • 1 points • 2025-09-26 06:16 UTC

Artist here. I think most artists are more concerned with the ethical and economic impacts of ai overuse more than the environmental impact, but that's included as a factor. Most artists probably do not understand the amount of resources that goes into any work they do. Art has never exactly been good for the environment.

However, the environmental impact of some ai companies should not be dismissed. X-Ai and OpenAi are both causing small environmental disasters in their locales (XAi has even committed an environmental crime). This is not a specifically-ai issue, but more needs to be done about holding large companies accountable for harming local people and wildlife.

u/ABigChungusFan • 1 points • 2025-09-25 16:53 UTC

Local water displacement.

u/recoverygarde • 1 points • 2025-09-25 14:23 UTC

I would be curious how you calculated the digital art bit as an iPad doesn't use as much energy nor has the same amount of resources needed to create as a gaming PC. Those specifics are important to clarify

u/ClothesPristine7428 • 1 points • 2025-09-25 22:08 UTC

I did a computer running the average software, in my revised version ill be sure to include the varition in devices and software